

Two Case Studies

Constructing Insurable Risk Portfolios

Colombia Workshop

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Two Risk Retention Case Studies

- The Wisconsin Property Fund case considers an insurance risk pool of fund members - each needs to manage up to $p = 6$ risk types.
 - Exhibits many features found in a typical general insurance company.
 - Extensive fund loss histories are available.
- The Australian National University (ANU) case considers a portfolio of $p = 15$ risks comparable to that commonly encountered by risk managers at large corporations.
 - Empirical historical data are available for high frequency risk types, but. . .
 - Most risk types are low frequency lines - little data available.
 - Focus - sensitivity of optimal risk retentions to changes in assumptions.

Wisconsin Property Fund

- Underlying data are longitudinal - a cross-section of pool members observed over time.
- Pool members differ by
 - type (e.g., school, county, and other types of governmental units),
 - size (measured by exposure variables), and
 - other characteristics (e.g., presence of sprinkler systems)
- There are $p = 6$ risk types of interest:
 - Building and Contents (BC), Construction Equipment (CE), and four types of auto (A1-A4)
- Parameter model estimation already done in Frees, Lee, and Yang (2016) study
 - Used 2006-2010 data to fit a Tweedie model.
 - Covariates from 2011 are used to generate a forecast distribution
- We created a multivariate forecast distribution of $p = 6$ risks for each of the $n = 1,098$ pool members.

1. Risk Retention for a Specific Member

- Work with Member 2, Ashland County.

Table 1: Forecast Loss Distribution for Member 2 (000s of USD)

	BC	CE	A1	A2	A3	A4
Minimum	0.00	0.00	0.00	0.00	0.00	0.00
Median	0.00	0.00	5.47	4.48	3.25	0.00
Mean	55.75	10.83	14.66	23.90	8.14	5.77
95th Percentile	298.26	64.63	57.09	102.96	30.99	31.00
99th Percentile	581.21	118.66	91.55	168.22	48.79	56.13
Mean Percent of Total	46.83	9.10	12.31	20.08	6.84	4.85

The sum of means over the six risks is 119.06.

Existing Customer of a Business Owners Policy

- A business owner and an agent of a commercial insurer seek to determine retention limits for each of several risks covered by the business owners policy.
 - They identify the risks where protection is desirable, e.g., building and contents, auto and on.
 - As an existing customer, the insurer has available data from the customer and related businesses that can be used to develop predictive models for the distribution of each risk.

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- The goal is to find the best set of upper limit parameters u_j that minimize the uncertainty of retained risks subject an upper bound on the risk transfer cost, RTC_{max} .
- This illustration uses the expected shortfall to summarize the uncertainty and fair costs for the risk transfer cost.
 - The optimization algorithm is run 11 times for different values of RTC_{max} .

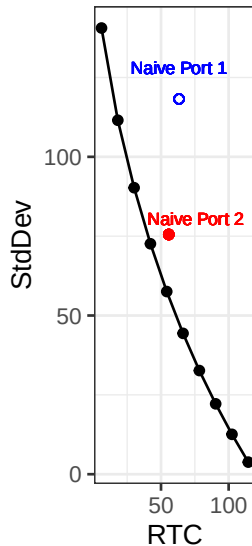
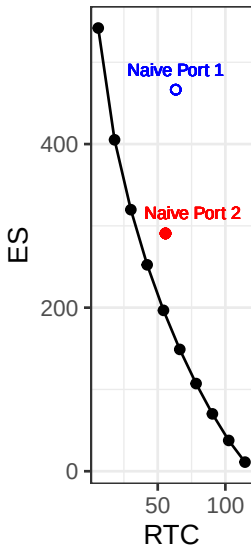
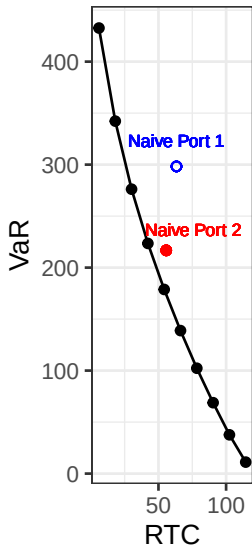
Optimization Results for Member 2

- They continue their discussion by examining the outputs from the algorithm that provides a menu of retention options organized in terms of increasing costs.

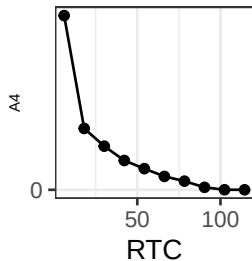
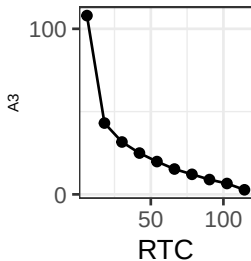
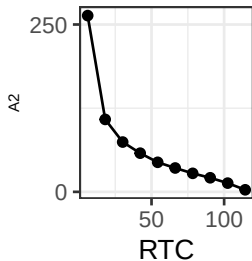
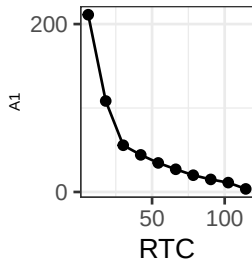
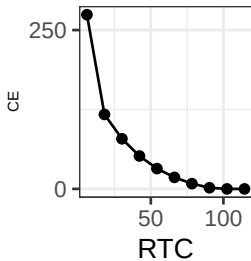
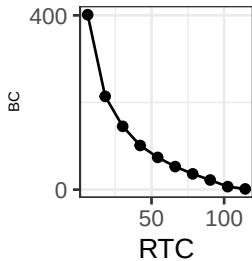
Table 2: Fund Member 2 Excess of Loss Optimization

RTC_{max}	RTC	VaR	ES	StdDev	u.BC	u.CE	u.A1	u.A2	u.A3	u.A4
115	115	11	11	4	1.7	0.0	3.6	3.1	2.8	0.0
102	102	38	38	13	6.9	0.0	11.1	13.0	6.6	0.0
90	90	69	70	22	22.1	1.8	14.9	21.0	8.9	1.8
78	78	102	107	33	36.2	8.3	20.0	27.6	12.1	6.4
66	66	139	149	44	52.9	18.2	27.0	35.6	15.3	9.9
54	54	179	197	58	74.0	32.0	34.6	44.2	19.8	15.6
42	42	223	252	73	101.4	51.8	44.2	57.8	25.0	21.6
30	30	276	320	90	145.3	78.9	55.6	74.4	31.6	32.2
18	18	342	405	112	214.1	117.2	108.4	108.2	43.1	45.3
6	6	433	542	141	401.5	274.2	211.2	263.3	108.0	128.7

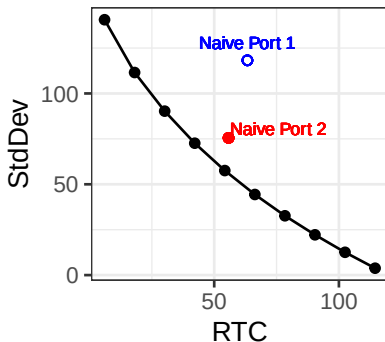
Efficient Frontier of Retained Risk Measures versus RTC



Upper Limits versus Risk Transfer Costs



RTC_{max}	54
$u.BC$	74
$u.CE$	32
$u.A1$	34.6
$u.A2$	44.2
$u.A3$	19.8
$u.A4$	15.6



- The graph compares the efficient frontier to some naive alternatives, created by fully insuring and retaining selected risks
- Select a specific $RTC_{max} = 54$, (000s of USD)
- The table shows optimal retention limits, a starting point for risk retention discussions

Australian National University (ANU) Case

- To illustrate its size, in 2020 it enjoyed an asset base of about 4.6 billion AUD
- To illustrate its complexity, its financial statements summarize many risks including property damage, general liability, cyber security, and so forth.
 - There are 15 risks in total, the “usual suspects.”
 - It pays about 25 million AUD per year in insurance premiums.
- In the absence of risk transfer agreements, ANU has responsibility for

$$S = X_1 + \cdots + X_{15}.$$

- In 2020 ANU suffered a major property loss (250 million AUD) meaning that its risk appetite has been subject to renewed scrutiny
 - This suggests that we will need to employ models that allow for long-tail distributions that permit very high values of outcomes with not insignificant probabilities.

Class of Insurance	Insurer	Deductible	Limit	Premium
Property	London Syndicate and Others	5,000,000	1,000,000,000	23,564,759
General and Products (G & P) Liability	Newline	100,000	20,000,000	340,000
G & P Umbrella Liability	Liberty	20,000,000	50,000,000	27,500
G & P 1st Excess Liability	QBE	50,000,000	100,000,000	27,500
G & P 2nd Excess Liability	Chubb	100,000,000	150,000,000	17,500
G & P 3rd Excess Liability	CGU	150,000,000	200,000,000	16,000
G & P 4th Excess Liability	Zurich	200,000,000	250,000,000	20,000
Cyber	London	250,000	2,000,000	75,721
Crime	AIG	100,000	20,000,000	100,000
Employment Practices Liability	AIG	100,000	2,000,000	84,000
Expatriate	Chubb			11,676
Group Personal Accident	Chubb	Various	As per schedule	104,920
Corporate Travel	Chubb	Various	As per schedule	75,000
Professional Indemnity	Newline	100,000	\$20m / \$40m	130,000
Medical Malpractice	Newline	100,000	\$20m / \$40m	
Clinical Trial	Newline	2,500	\$20m	
Statutory Liability	Berkley Insurance Australia (SUA)	\$1,000 / \$15,000	1,000,000	8,360
Motor Vehicle	Vero	1,000	As per schedule	84,700
Marine Cargo	Richard Oliver (QBE)	5,000	5,000,000	6,127
Marine Hull	Richard Oliver (QBE)	150	5,000,000	11,552
TOTAL				24,407,255

Optimizing the ANU Risk Portfolio

- Because of its size, the algorithm performs poorly when including the Property Risk
 - Instead, assume that the risk owner transfers all property risks in excess of 5000, in thousands of AUD.
 - The portfolio of ANU's retained risks is

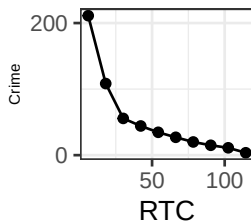
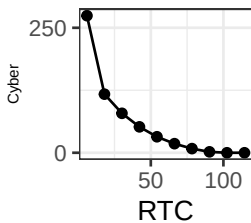
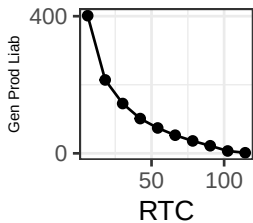
$$S_{Prop}(u_2, \dots, u_{15}) = \min(X_1, 5000) + \min(X_2, u_2) + \dots + \min(X_{15}, u_{15}).$$

- The goal is to determine the optimal values of the upper limits $\{u_2, \dots, u_{15}\}$.
 - Property is still included in the risk portfolio but not part of the potential risk transfer costs.
- The motivation is that that this treaty will be negotiated separately from other risks
- The 5000 level is the one currently employed by ANU.
 - For sensitivity analysis, also examined a fixed level of 1000.

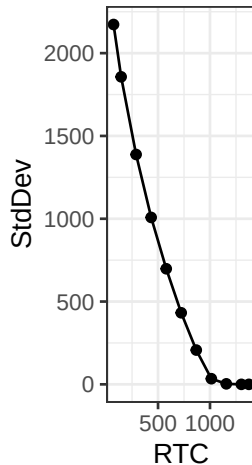
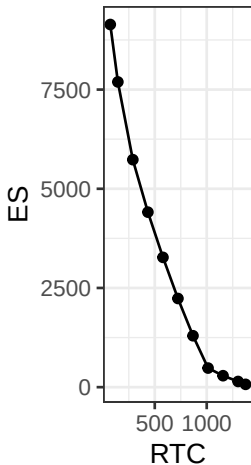
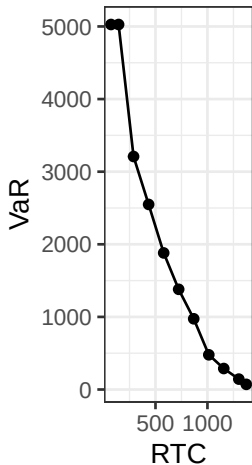
Table 3: ANU Optimization over Levels of Maximal Risk Transfer

RTCmax	RTC	VaR	ES	Std Dev	u2	u3	u4
1374	1374	5069	5069	2030	0	0	0
1302	1302	5141	5141	2030	0	0	0
1157	1157	5287	5287	2031	0	0	0
1013	1013	5476	5478	2038	15	0	0
868	868	5870	6286	2124	140	170	86
723	723	6172	7195	2233	334	347	262
579	579	6532	8169	2372	640	639	542
434	434	6825	9205	2554	1119	1167	1117
289	289	7332	10369	2811	1910	3166	3213
145	145	7507	11717	3139	5244	5582	5614
72	72	8089	12799	3384	9217	7242	7372

ANU Optimization - First Three Retention Levels



ANU Optimization Results - Efficient Frontier



Next Steps

- The Wisconsin Property Fund case exhibits many features found in a typical general insurance company.
- The Australian National University (ANU) case with risks comparable to that commonly encountered by risk managers at large corporations.
- In the next block of workshop time, pick one and explore the code available in the Jupyter notebooks.
- Next, go to the online book for additional code that you can cut-and-paste to explore additional features.
- Finally, think about how you might use this framework in your own investigations (use Gemini notebook to help you explore!)

Selected Reference

Frees, Edward W, Gee Lee, and Lu Yang. 2016. "Multivariate Frequency Severity Regression Models in Insurance." *Risks* 4 (1): 4. <https://doi.org/10.3390/risks4010004>.